

ASGA™

Autonomous Systems Governance Architecture

Official Autonomous Operational Governance Architecture for Autonomous Systems, AI Agents, Robotics, Autonomous Vehicles, Industrial Automation, Critical Infrastructure, Human-AI Environments, and Future Autonomous Ecosystems

ASGA™ serves as the official OOF® autonomous operational governance architecture used for autonomous operational governance, autonomous operational state identification, autonomous operational diagnostics, autonomous architecture navigation, and autonomous operational problem placement across AI systems, autonomous systems, robotics, industrial automation, cyber-physical systems, Human-AI environments, and future autonomous ecosystems.

Autonomous Systems Governance Architecture™ defines the governance architecture required to establish, govern, validate, coordinate, supervise, and preserve trustworthy autonomous operation throughout the complete autonomous operational lifecycle.

ASGA™ consists of 10 Core Parent Standards forming the primary governance backbone of autonomous operational governance.

Additional compatible standards, bridge standards, and specialized autonomous governance standards may extend the architecture as new autonomous operational spaces emerge. However, the standards listed below form the official core governance structure of ASGA™.

Canonical Definition

Autonomous Systems Governance Architecture™ is the governance architecture that defines the structural conditions under which autonomous operation remains identifiable, authorized, executable, constrained, trustworthy, coordinated, accountable, compatible, evolvable, and continuously governable across complex autonomous operational ecosystems.

Core Question

How can autonomous systems remain continuously governable while operating independently throughout the complete autonomous operational lifecycle?

Governance Flow

Identity → Authority → Execution → Constraints → Risk & Trust →
Coordination → Escalation → Accountability → Compatibility →
Evolution

Core Parent Standards

1. Autonomous Identity Standard (AIS)

Governed Space: Autonomous Operational Identity

Core Question: Who is the autonomous operational actor?

2. Autonomous Authority Standard (AAS)

Governed Space: Autonomous Operational Authority

Core Question: Who has authority to perform autonomous operations?

3. Autonomous Execution Standard (AES)

Governed Space: Autonomous Execution

Core Question: What autonomous actions may be executed?

4. Autonomous Constraint Standard (ACS)

Governed Space: Autonomous Operational Constraints

Core Question: Which operational constraints may never be violated?

5. Autonomous Risk & Trust Standard (ARTS)

Governed Space: Autonomous Operational Risk & Trust

Core Question: What level of operational risk and trust is acceptable?

6. Autonomous Coordination Standard (ACOS)

Governed Space: Autonomous Operational Coordination

Core Question: How do autonomous systems coordinate operations?

7. Autonomous Escalation Standard (AESS)

Governed Space: Autonomous Operational Escalation

Core Question: When and how should autonomous operation escalate?

8. Autonomous Operational Accountability Standard (AOAS)

Governed Space: Autonomous Operational Accountability

Core Question: How is accountability preserved throughout autonomous operation?

9. Autonomous Compatibility Standard (ACPS)

Governed Space: Autonomous Operational Compatibility

Core Question: How do autonomous systems remain operationally compatible?

10. Autonomous Evolution Standard (AEVS)

Governed Space: Autonomous Evolution **Core Question:** How may autonomous systems evolve while remaining governable?

Compatible OOF® Architectures

ASGA™ interoperates with multiple OOF® architectures governing complementary operational spaces.

GOA™ — Governance Architecture provides the constitutional governance layer defining how governance itself is established, delegated, validated, protected, and coordinated.

ORA™ — Operational Reality Architecture governs operational reality, operational validity, operational authority, operational evidence, and operational continuity throughout execution.

AGA™ — Accountability Governance Architecture governs accountability relationships, ownership, delegation, attribution, and responsibility across autonomous operational environments.

AIG® — AI Governance Architecture governs AI behavior, behavioral integrity, explainability, transparency, and trustworthy AI decision-making.

CLIA® — Cognitive Governance Intelligence Architecture governs cognition, reasoning, interpretation, cognitive integrity, and cognitive decision processes supporting autonomous operation.

MGIA™ — Memory Governance Intelligence Architecture governs memory continuity, memory integrity, traceability, retention, and governed operational memory throughout autonomous execution.

RIS™ — Runtime Integrity Standard provides the runtime integrity layer supporting trustworthy execution across autonomous operational environments.

Architectural Position

ASGA™ governs the complete autonomous operational lifecycle.

The architecture progresses through:

Autonomous Identity → Autonomous Authority → Autonomous Execution → Autonomous Constraints → Autonomous Risk & Trust → Autonomous Coordination → Autonomous Escalation → Autonomous Operational Accountability → Autonomous Compatibility → Autonomous Evolution

Together these standards govern how autonomous systems establish identity, receive authority, execute operations, remain within operational constraints, manage operational risk, coordinate with other systems, escalate exceptional situations, preserve accountability, maintain compatibility, and evolve while remaining continuously governable.

Canonical Principle

ASGA™ does not govern what autonomous systems are intended to achieve. ASGA™ governs whether autonomous operation remains identifiable, authorized, constrained, trustworthy, coordinated, accountable, compatible, evolvable, and continuously governable throughout the complete autonomous operational lifecycle.

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Canonical Source

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